nature energy

Article

https://doi.org/10.1038/s41560-026-01992-0

Sorption-driven dissolution refrigeration cycle with thermal storage

Received: 1 May 2025

Accepted: 22 January 2026

Published online: 10 March 2026

Check for updates

Si Wu[1,2], Kaiyue Tang[1,2], Xiao Zhang[1], Haoyuan Sang[1], Ruxue Du[1], Pengfei Wang[1], Jiaxing Xu[1], Ruzhu Wang[1] & Tingxian Li[1]

Developing energy-efficient and environmentally friendly refrigeration technology is highly desired to tackle climate change. Emerging caloric effect-based cooling technologies such as magneto-, electro- and ionocaloric effects are promising but suffer from large driving field strengths, low adiabatic temperature change or insufficient power density. Here we propose a sorption-driven dissolution refrigeration cycle with an extendable thermal storage function. Theoretical and experimental results show larger adiabatic temperature change compared with solid-state caloric effects by using medium- or low-grade heat (80–150 °C) as the cyclic driving energy. We demonstrated the viability of a practical system using such a cycle, with experimental results showing adiabatic temperature change of 37 K and a minimum cooling temperature of −25.4 °C. Importantly, the sorption process allows the heat storage and flexible cold and heat supply to adapt to the diverse and complex application scenarios. This work shines light on the exploitation of renewable energy for efficient cooling and heating.

Refrigeration and air conditioning account for approximately 20% of global electricity consumption, which is rapidly increasing with the ongoing global warming and population growth[1]. Vapour compression cooling technology dominates daily cooling applications and relies heavily on hydrofluorocarbons that possess global warming potentials (GWPs) far higher than CO_2 (ref. 2). Therefore, the development of energy-efficient and environmentally friendly refrigeration technologies has become an urgent priority. Emerging cooling technologies based on caloric effects, such as the magnetocaloric effect[3–5], electrocaloric effect[6–8], elastocaloric effect[9–11] and barocaloric effect[12–14], have been extensively explored as potential alternatives to vapour compression cooling technology owing to zero GWP. However, these solid-state cooling methods typically require large driving field strengths but yield small adiabatic temperature changes and low power density outputs compared with conventional vapour compression cycles[15].

The recently reported ionocaloric cooling cycle based on the ionocaloric effect, defined as the thermal response to a changing ionic environment surrounding a solid phase, has demonstrated a remarkable adiabatic temperature change of 25 K even under a low voltage strength of ~0.22 V by using an ethylene carbonate–sodium iodide (NaI–EC) working pair. However, the cooling power density achieved was only 5.75–180 W L^{-1} and a tradeoff emerged between relative Carnot efficiency and power density[16]. Such a low cooling power is due to the slow kinetics of mixing two solid materials in the cooling stage, presenting a long-term challenge for improving power density.

Alternatively, mixing solids with liquid substances (that is, the dissolution process) is expected to have fast kinetics and cooling power if an endothermic effect is yielded during the mixture. In this case, dissolving an endothermic salt into the water can generate the cooling effect due to the change in the ionic environment of water under a chemical field. Through this field, the solute transforms from a solid to a dissolved state. As the dissolved phase becomes thermodynamically more stable than the solid phase, it absorbs heat to realize refrigeration[17–19]. On the basis of this principle, ancient societies developed the saltpeter ice-making technique using the endothermic effect accompanied by the dissolution of potassium nitrate into water[20].

[1]Institute of Refrigeration and Cryogenics, MOE Engineering Research Center of Solar Power & Refrigeration, School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai, China. [2]These authors contributed equally: Si Wu, Kaiyue Tang. e-mail: Wusi716@sjtu.edu.cn; Rzwang@sjtu.edu.cn; Litx@sjtu.edu.cn

Recently, a solution cooling cycle using an ammonium nitrate–water (NH_4NO_3–H_2O) working pair was constructed and solar thermal energy serves as the driving force to separate the salt and water for thermal distillation regeneration[17]. Although a cooling power of 191 W m^{-2} and a temperature change of 26.4 K were achieved, the overall system efficiency remained low due to the large discrepancy between the benefit and the cost, that is, the dissolution enthalpy is smaller than the phase change enthalpy required for water evaporation during the regeneration of salt. Consequently, approximately 80% of solar energy has to be wasted as condensation heat. Very recently, a continuous dissolution refrigeration system (called liquid-state dipolarcaloric) using NH_4NO_3–H_2O and KNO_3–H_2O working pairs has been reported, employing electrodialysis for salt–water separation[21]. While it demonstrates a relatively high coefficient of performance (COP) and cooling power, the system still faces challenges such as the consumption of high-grade electrical energy and the use of explosive nitrate salts.

Here, we propose a sorption-driven dissolution refrigeration (SdDR) cycle, in which a sorption process is introduced into the dissolution refrigeration cycle to provide the driving force for regenerating the dissolved endothermic salt through reduced-pressure distillation and simultaneously serves as a heat recovery unit by exothermally adsorbing the distilled gas. This cycle utilizes medium/low-grade heat (80–150 °C) as the cyclic driving energy and employs a safe working pair for operation. Basically, the recovered heat is from the desorption heat for the regeneration of sorbent accompanied by the release of sorbates (that is also the distilled gas). Consequently, the coupling of multiple processes in the SdDR cycle offers extensive opportunities for multifunctional applications, for example, the demand-oriented regulation of solar energy for cooling and heating. When operated by intermittent means, this cycle can be powered for dissolution refrigeration by harnessing abundant solar energy, while the inverse process can provide thermal energy on demand. This temporal decoupling of cold and heat outputs offers an effective solution to the inherent mismatch between cooling and heating demands and solar energy availability (Supplementary Figs. 1–3). As a proof of concept, we construct the SdDR cycle using water as the solvent and the cost-effective hygroscopic salt as the sorbent. The results show that the endothermic effect demonstrates a 37 K adiabatic temperature change and a −25.4 °C minimum cooling temperature by employing potassium thiocyanate as the endothermic salt. The overall COP reaches up to 1.13 when the system is operated in the mode of alternate cooling and heating.

Concept of the SdDR

The SdDR system comprises two materials (that is, a sorbent and an endothermic salt) that share the same medium (that is, water in this proof of concept). Similar to the ionocaloric effect[16], the endothermic salt in the SdDR cycle experiences two processes: mixing with water to form a salt solution and separating from water to revert to endothermic salt (Fig. 1a). The mixing process is accompanied by substantial heat absorption from the surroundings due to the combined function of hydration energy and lattice energy, thereby achieving the cooling effect as depicted by the temperature and entropy changes in stages 4–5 (Fig. 1b). In the separation process, the dry sorbent adsorbs water in the form of vapour from the solution to regenerate the endothermic salt, accompanied by substantial heat release due to the affinity interaction between water molecules and sorbent, as illustrated in stages 6–9 (Fig. 1b). In principle, the temperature required for regeneration of the endothermic salt can be tuned from the solution's freezing point up to ambient temperature, offering an extensive working temperature range for the SdDR cycle. This is indicated by the interval between stages 5 and 6 (Fig. 1b). The driving force behind the endothermic salt–water separation and vapour sorbent sorption processes is the difference between the two equilibrium pressures of dry sorbent and solution (Supplementary Fig. 4). Specifically, the transition of water molecules from a liquid to a gaseous state increases the solution pressure to its saturated vapour pressure, while the sorption process reduces the pressure in the vicinity of the sorbent as water molecules transform from the gaseous to the adsorbed state, creating a pressure difference that drives the endothermic salt–water separation process. For the regeneration of dry sorbent, thermal energy is needed to drive the desorption of water from the sorbent to yield vapour that is then condensed into liquid water for the subsequent dissolution refrigeration (Fig. 1a). This process is illustrated in stages 1–4 (Fig. 1b). This heat input is the only energy consumed to power the overall cycle and it could be from ubiquitous medium- and low-grade heat sources, such as solar thermal, industrial waste heat and so on. In comparison with the conventional water-based thermochemical heat storage system, this novel approach offers two notable advantages: (1) the salt solution has a lower freezing point than pure water, enabling the sorption process for heat release even at very low ambient temperatures, and (2) it yields the extra cooling effect in the process of energy storage to rightly match the intensity of solar irradiance and cooling demand (Fig. 1c and Supplementary Fig. 5).

Selection of working pairs

For the endothermic salt, the adiabatic temperature change and the refrigeration capacity are two key performance indicators, primarily determined by the enthalpy change of dissolution and solubility in water. Ideal endothermic salts should have a high dissolution enthalpy change and solubility, enabling substantial cooling with minimal material usage. NH_4NO_3, NH_4NO_2, NH_4CNS, KSCN and $CO(NH_2)_2$ were initially selected as potential candidates owing to their relatively high cooling capacities and gravimetric energy densities, which are comparable to the solid–liquid transition enthalpies (~200 kJ kg^{-1}) of many organic and inorganic materials (Supplementary Fig. 6). Several commonly used hygroscopic salts were selected as the vapour sorbent to pair with the endothermic materials for further evaluating the performance of the whole cycle. When paired with the salt sorbent, NH_4NO_3 and NH_4NO_2 exhibit higher cooling power under a given thermal input power (Fig. 2a), but both of them suffer from instability and a risk of explosion, limiting their practical applications. Among the remaining three endothermic salts, KSCN demonstrates the highest temperature change and cooling power (Supplementary Fig. 7a). In addition, the eutectic point of the KSCN aqueous solution reaches an extraordinary −47 °C, theoretically enabling the evaporation at extremely low ambient temperatures to provide water vapour for the sorption process with heat release (Supplementary Fig. 7b).

For the sorption process, the affinity of the salt sorbent with vapour affects the regeneration of endothermic material and the heat release temperature of the sorption process. Compared with other salt sorbents, lithium bromide (LiBr) exhibits the highest output temperature but suffers from a low energy density, whereas lithium chloride (LiCl) offers a relatively high output temperature and high energy density, indicating its strong sorption capacity (Fig. 2a,b and Supplementary Fig. 8). On the basis of these considerations, the KSCN–water–LiCl working pair was selected to construct the SdDR cycle as it demonstrates a relatively high thermal storage density and cooling power for the system. The quantitative SdDR cycle using this working pair is depicted in Fig. 2c. Initially, LiCl·H_2O undergoes isosteric heating (Fig. 2c, a–b), followed by isobaric desorption and further heating to the heat source temperature (Fig. 2c, b–c), during which the desorbed vapour condenses to liquid water under the ambient heat sink (Fig. 2c, A). Then, the room-temperature liquid water mixes with KSCN forming an aqueous KSCN solution through an endothermic dissolution process that generates a cooling effect (Fig. 2c, A–B). During the transition stage, the salt sorbent and the solution are separated physically and the temperatures of them both vary with surrounding temperature (Fig. 2c, c–d–e and B–C), for example, −10 °C in Fig. 2c. Accordingly, their equilibrium pressures change according to the Clausius–Clapeyron equation and the pressure difference between them is

https://doi.org/10.1038/s41560-026-01992-0

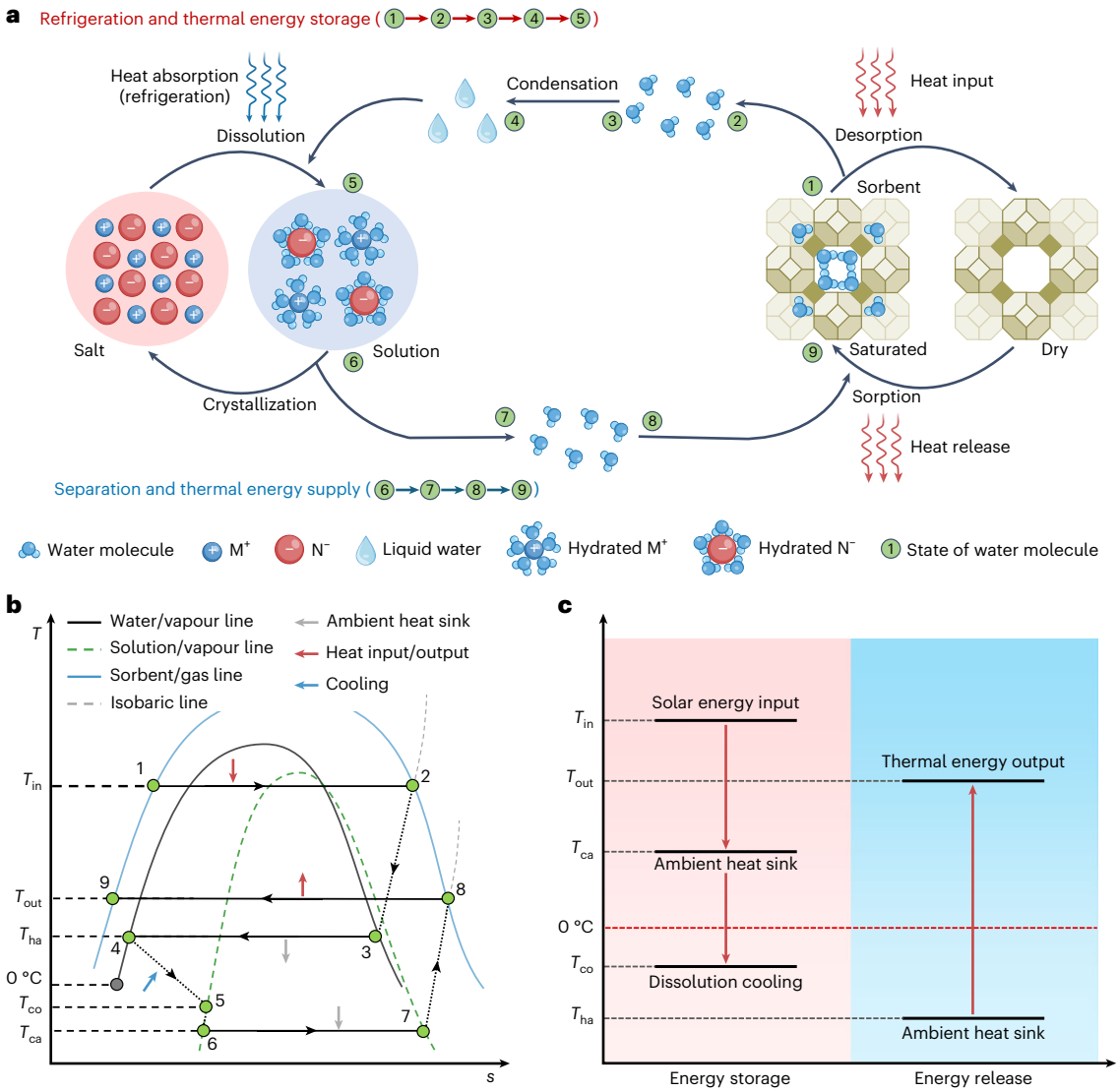

Fig. 1 | Concept and fundamentals of SdDR. a, A schematic diagram of the SdDR cycle. The numbers within the green circles correspond to the states of water molecules throughout the cycle, as depicted in **b**. M^+ and N^- represent the cations and anions, respectively. **b**, The temperature–entropy diagram of the SdDR cycle shows the state evolution of water molecules in the whole process. T_{in}, T_{out}, T_{ca}, T_{ha} and T_{co} represent the input and output temperatures of heat, the ambient temperatures in the process of cooling and heating, and the temperature of cooling, respectively. **c**, The energy change process in this work. The solar energy input is used for desorption and then converted into liquid water condensed by the desorption gaseous water and finally the liquid water mixes with the endothermic salt to produce a cooling effect. The environment is used as a heat sink and the water in the solution is absorbed into the sorbent to produce a heating effect.

created, which is able to drive the evaporation of solution and the flow of water vapour once the solution side and sorbent side are physically connected. The water vapour is adsorbed by the sorbent (Fig. 2c, e–f) and meanwhile, the sorption heat is released at more than 40 °C, which is considerable to satisfy heating demands in cold environment (−10 °C).

In practical conditions, the slow kinetics of pure hygroscopic salts can hinder the hydration–dehydration reaction, which has been a long-term challenge. Confining the salt within porous matrices such as metal–organic frameworks, aerogels or zeolite is a general solution[22–24]. Among these matrices, zeolite 13X possesses a relatively high vapour sorption affinity and has been used both as a physical sorbent and porous matrix (Supplementary Fig. 9). Therefore, we selected zeolite 13X to load LiCl for preparing composite sorbent, which exhibits enhanced sorption capacity, good cycling stability over 35 cycles and exceptional sorption strength, capable of adsorbing vapour at a relative pressure as low as 0.5% (Supplementary Figs. 10–12). It should be mentioned that there exist large amounts of alternative working pairs that are appropriate for different operation temperatures, as shown in Supplementary Note 1 and Supplementary Fig. 13.

Performance evaluation of dissolution refrigeration

The dissolution enthalpy of endothermic salts is a key metric to evaluate the energy density for cooling process. As shown in Fig. 3a, the dissolution enthalpy increases from 120 kJ kg_{salt}^{-1} to 180 kJ kg_{salt}^{-1} as the dissolution temperature varies from −10 °C to 30 °C (Methods and Supplementary Fig. 14), which can be attributed to the continuous weakening of hydration with temperature rising. To further explore the cooling temperature limit, the initial temperatures of the KSCN salt and water before mixing are adjusted to suppress the effect of sensible heat on the temperature drop. When the initial temperature of water is reduced from 20 °C to 0 °C, the temperature drop decreases only from 32.8 K to 25.8 K (Fig. 3b). In addition, the adiabatic

https://doi.org/10.1038/s41560-026-01992-0

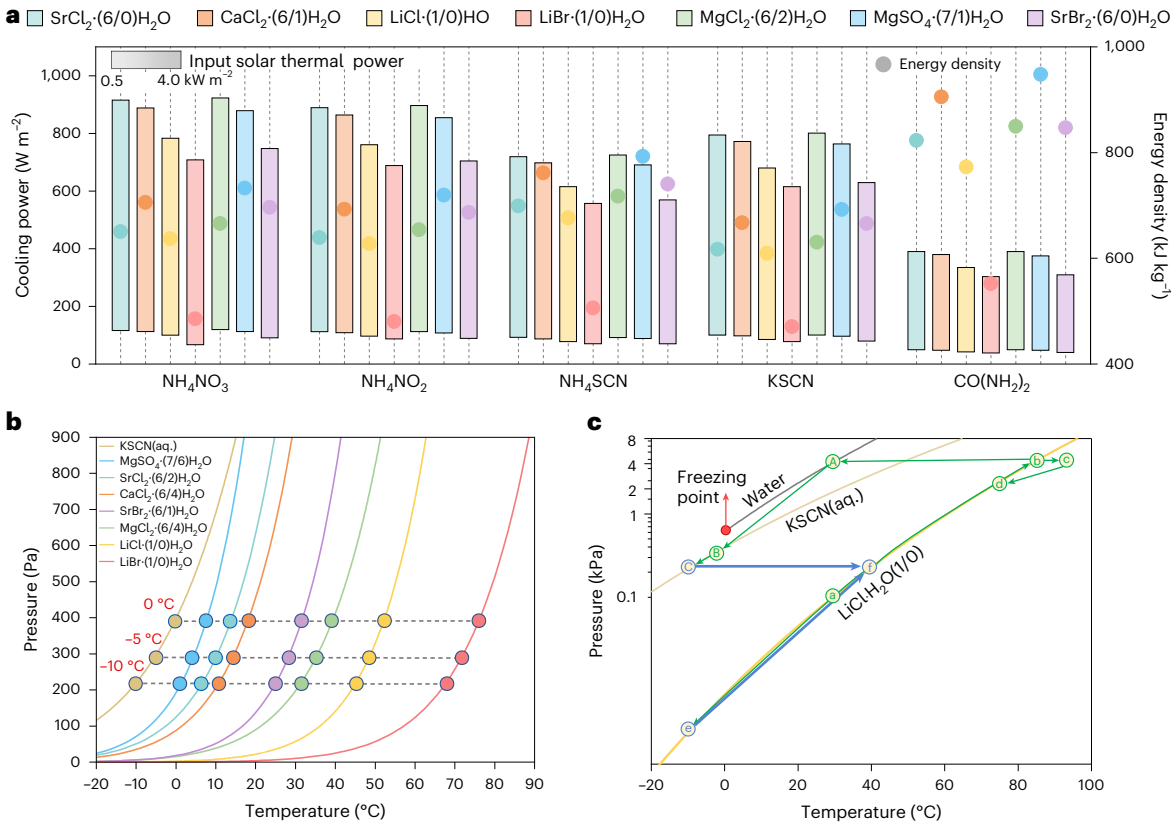

Fig. 2 | Characterization of endothermic materials and sorbents. a, The working performance of various working pairs. The coloured columns represent the cooling power under different solar thermal energy power inputs, while the coloured dots indicate the thermal storage density, with the unit mass referring to the total mass of all materials. The vertical dashed lines serve as visual guiding lines for each sorbent–salt working pair. For each working pair, the range of cooling power generated by salt dissolution process is linked with the input solar thermal power driving the dehydration process of salt hydrates through the generation rate of vapour. **b**, The comparison of the minimum heat release temperatures (or minimum sorption equilibrium temperature) for typical salt sorbents at different separation temperatures of saturated KSCN aqueous solution. **c**, The thermodynamic cycle in a Clausius–Clapeyron diagram, indicating the variations of pressure and temperature for the selected KSCN–water–LiCl working pair. The letters a–f represent the states of LiCl·H_2O/LiCl in the cycle. A–C represent the states of water and KSCN solution. The green lines represent the cooling process and surrounding temperature change (B–C and d–e). The blue lines represent the separation process.

temperature change of the KCSN–water system increases with the chemical potential (Supplementary Fig. 15). Surprisingly, the minimum temperature of the binary system reaches as low as −25.4 °C within 20 s by mixing water with KSCN particles at nearly 0 °C, demonstrating the applicability of the endothermic effect across various conditions (for example, spacing cooling, food storage and so on) (Fig. 3b and Supplementary Videos 1–5).

The multistage endothermic effect was also measured by incrementally adding water to the endothermic salt (Methods). The results indicate that the cooling effect remains nearly consistent without attenuation until the salt is fully dissolved, with a residual cooling effect persisting for the saturated solution owing to the contribution of the dilution enthalpy (Fig. 3c and Supplementary Fig. 16). As the quantity of water increases, the temperature drop decreases due to the increased sensible heat consumption by the preceding solution. Even so, a large temperature drop of 12.2 K is still observed for the third addition of water (Fig. 3c). These results certify the cooling capacity of the supersaturated salt solution where there still exists undissolved salt in the binary system. In addition, the average cooling power density reaches 195 W L^{-1} that is determined by the fluid heat-transfer method (Supplementary Fig. 16d). Different from the solid-state caloric and ionocaloric materials, the salt–water mixture is pumpable, enabling enhanced heat and mass transfer during both dissolution and regeneration. Figure 3d compares KSCN with other endothermic salts (NH_4NO_3 and KNO_3)[21] in five key dimensions: adiabatic temperature change, isothermal entropy change, energy density, solubility and safety. The adiabatic temperature change and isothermal entropy change of KSCN reach 37 K at 35 °C and 433.7 J kg^{-1} K^{-1} at 30 °C, respectively. Compared with NH_4NO_3 and KNO_3, KSCN not only exhibits comparable cooling performance, but also has the higher solubility and energy density (Supplementary Table 1). Furthermore, it is safer and more easily available than the explosive and controlled nitrate salts, while NH_4NO_3 continues to pose a threat to human safety and causes many accidents[25,26]. More information of adiabatic temperature change and isothermal entropy change regarding other caloric technologies and vapour compression technology is presented in Supplementary Tables 2–4.

Sorption-driven regeneration of the endothermic salt

To validate the feasibility of the sorption-driven regeneration of the KSCN–water solution, a homemade prototype containing the dry LiCl@Zeolite13X composite sorbent and the saturated KSCN–water solution was built and their temperature evolutions were monitored to act as the indicator for the regeneration process (Methods). The two components are initially separated by a valve installed between them. Upon connecting the two ends to initiate the sorption process, a rapid temperature rise on the sorbent side (T_{hot}) and, meanwhile, a sharp temperature drop on the solution side (T_{cold}) are observed owing to the exothermic sorption and endothermic evaporation, respectively (Fig. 4a). This indicates the success in separating the binary KSCN–water

https://doi.org/10.1038/s41560-026-01992-0

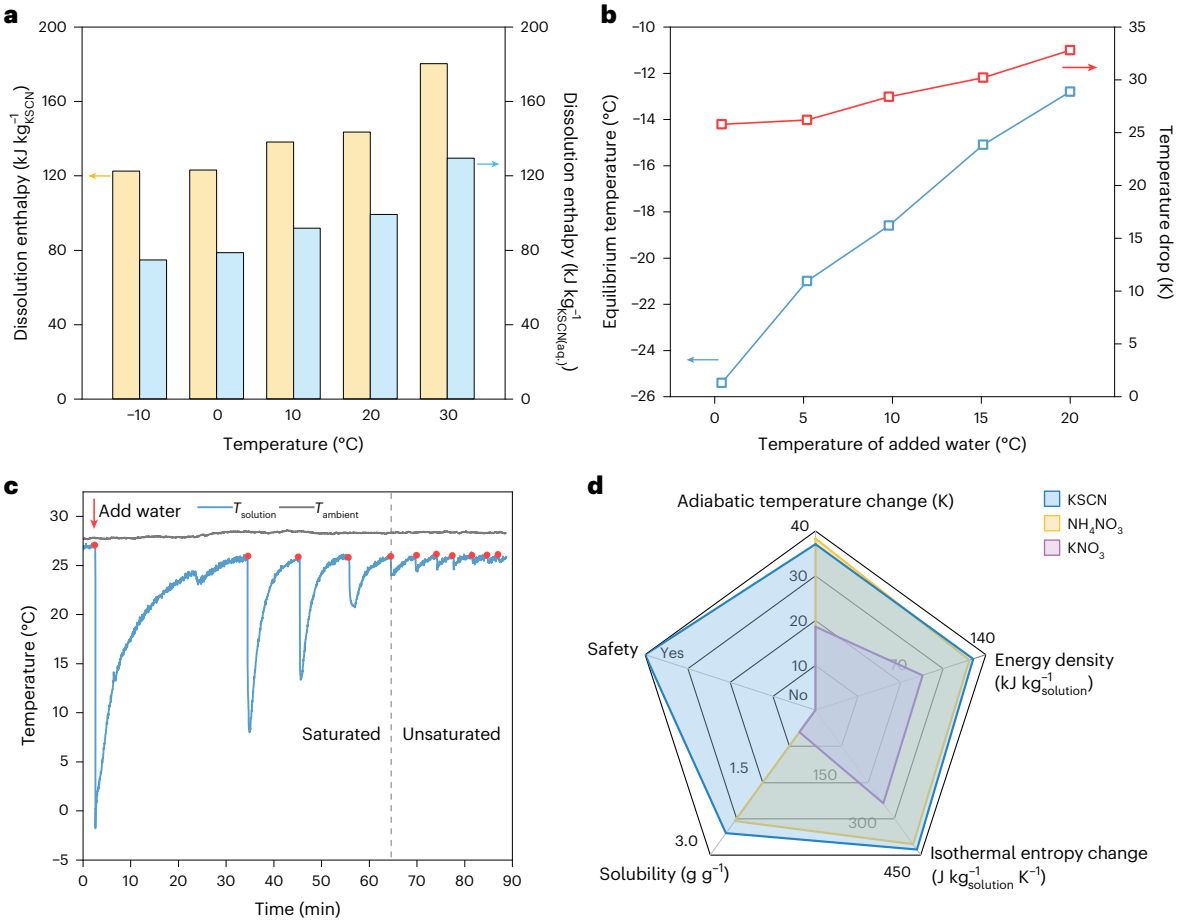

Fig. 3 | The endothermic effect of the KSCN–water system. a, Dissolution enthalpy at different temperature calculated based on different mass levels, including KSCN and KSCN(aq.). **b**, The refrigeration temperature limit of the KSCN–water system and the influence of the initial temperature of water on its temperature change. **c**, The multistage dissolution refrigeration effect. The red dots represent the addition of water. **d**, A comparison of the material characteristics of endothermic salts in five dimensions: adiabatic temperature change at 35 °C, isothermal entropy change at 30 °C, energy density at 30 °C, solubility at 30 °C and safety. The specific data are given in Supplementary Table 1.

system or the regeneration of KSCN. The maximum temperature deviations from the ambient temperature, T_{heated} and T_{cooled}, reach 23 K and 32 K, respectively. The simultaneous cold and heat output endows this system with versatile application potential. Apart from the alternate cooling and heating operation mentioned above, both continuous heating and cooling operation can be achieved by regulating the working temperature or pressure of each component (Supplementary Fig. 17). Particularly, for a low ambient temperature heat supply, the sorption heat release under a low evaporation temperature is a main concern.

To further explore the climate adaptability of the system, the cold end (solution side) is controlled at different temperatures ranging from 0 °C to −13 °C. It is found that the hot end (sorbent side) can still achieve a maximum temperature of 43.2 °C at −13 °C evaporation temperature, demonstrating the broad operational range of the separation process and its superior climate adaptability across extreme conditions (Fig. 4b). However, as the cold-end temperature becomes lower, the separation becomes more challenging owing to the reduced driving pressure difference. Accordingly, the maximum attainable temperatures of the sorbent side will drop. Therefore, selecting a sorbent with a strong affinity for water vapour is essential to ensure the useful heat output at an appropriate temperature. For the KSCN–LiCl@Zeolite13X working pair, the maximum hot-end temperature varies from more than 40 °C to nearly 60 °C at the cold-end temperature spanning from −13 °C to 0 °C (Fig. 4b), which is sufficient for various applications such as space heating, hot water supply and so on. Changing the initial temperature of sorbent side from 10 °C to 40 °C has minimal impact on its thermal response characteristics while maintaining the cold end at −5 °C, demonstrating a stable temperature evolution trend across diverse thermal energy utilization scenarios. The maximum temperature rise only decreases slightly from 20 K to 18 K owing to the reduced initial driving force (Fig. 4c). As for the thermal energy storage density (TESD), the maximum heat output quantity is 763 kJ kg$^{-1}_{sorbent}$, calculated by the actual heat exchange capacity of heat transfer fluid (Supplementary Fig. 18), which is lower than the maximum theoretical value calculated by thermogravimetric and differential scanning calorimetry (Supplementary Fig. 19). This deviation is mainly due to the inevitable heat loss and the incomplete conversion of sorbent. Even so, the actual thermal storage density remains higher than that of the latent heat storage (generally <300 kJ kg^{-1}) for the same operation temperature range. The energy density for both heating and cooling is presented in Supplementary Table 5. It should be noted that the separation process is continuous in theory until the sorbent reaches saturation or water is completely separated from the solution, which is dominated by the persistent driving pressure differential across the system. This demonstrates the potential for complete separation by optimizing the ratio of sorbent and solution. The material's sorption capacity, the decisive factor of separation performance, is governed by the relative pressure around the sorbent, which is determined by the temperature difference between the solution and the sorbent. Based on the temperature of the solution and sorbent, the theoretical

https://doi.org/10.1038/s41560-026-01992-0

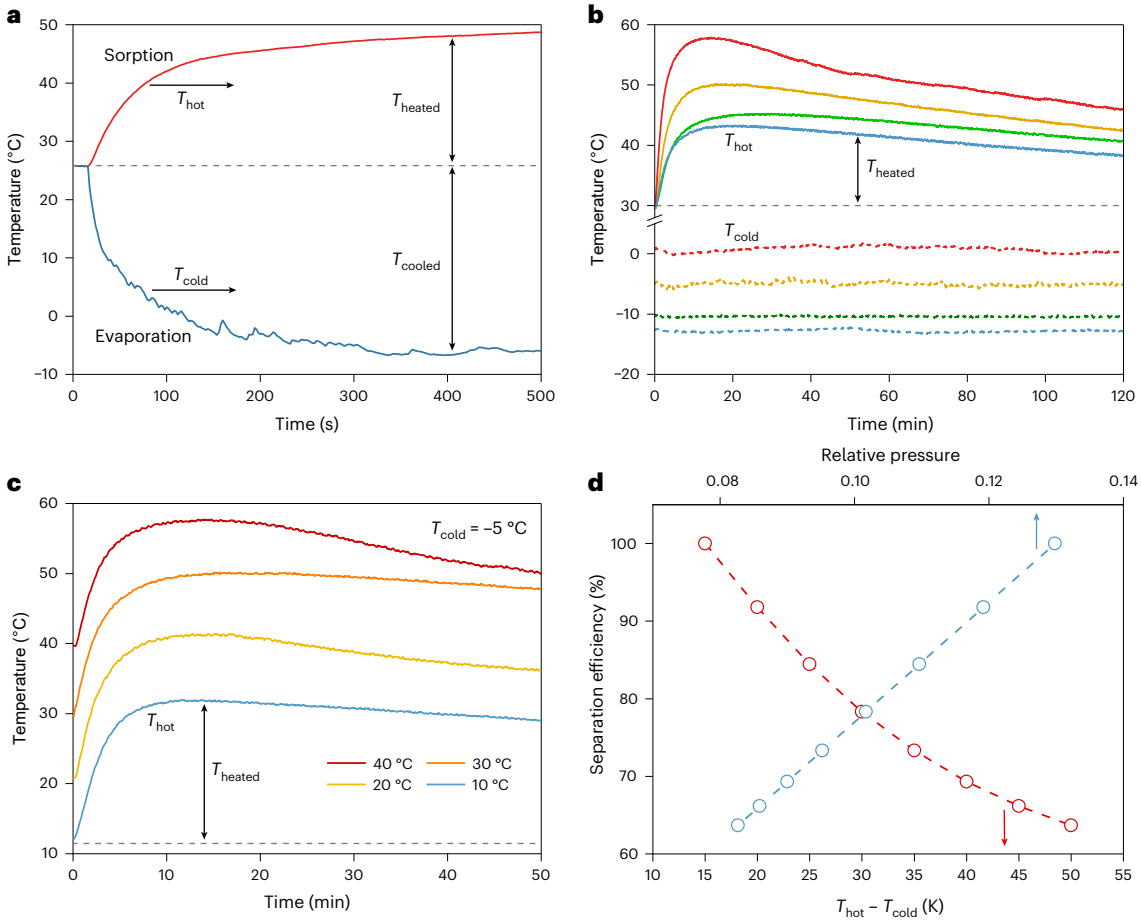

Fig. 4 | Characterization for separation of the KSCN–water binary system and sorption thermal energy output. a, The temperature evolution of the hot end (sorption side) and the cold end (solution side) during the sorption-driven separation of the KSCN–water binary system. **b**, Temperature evolutions of the sorption side under different evaporation temperatures of the saturated solution. The temperature evolutions of the sorption side (T_{hot}, corresponding to full lines) under different evaporation temperatures of the saturated solution (T_{cold}, corresponding to dotted lines). Each T_{hot} line corresponds to one T_{cold} line according to the line colour. **c**, Temperature evolutions of the sorption side at different initial temperatures for a given evaporation temperature of −5 °C. **d**, The theoretical separation efficiency of KSCN–water for different operating conditions. The temperature difference between the sorbent side and the solution side is 15 K in the case of complete separation. The relative pressure is defined as the ratio of the saturated vapour pressure of the solution at a given temperature to the saturated vapour pressure of water vapour at the temperature on the sorption side.

separation efficiency reveals remarkable system performance and reliability, with separation efficiency maintaining 50% retention even under a temperature difference of 50 K (Fig. 4d and Methods).

The whole cycle of SdDR

The whole SdDR cycle is performed in a homemade prototype, which comprises two half cycles and yields two benefits, that is, heat-powered refrigeration with simultaneous thermal storage and sorption heat supply, as shown in Fig. 5a. Three main components named the thermal storage tank, the condenser and the refrigeration tank in the prototype are isolated by vacuum valves (Supplementary Fig. 20). The temperature evolution throughout the entire cycle process is recorded in Fig. 5b.

In the charge stage, a high-temperature liquid at 120 °C is used as the heat source to desorb vapour from the composite sorbent. The rapid influx of high-temperature vapour causes the condenser temperature to rise sharply at first, then stabilize at a constant value as the heat from condensation balances the heat dissipated (Fig. 5b). Subsequently, the condenser temperature decreases to the ambient temperature, indicating that the desorption process is finished, with thermal energy stored in the form of sorption potential[24]. Then, by opening the valve between the condenser and the refrigeration tank, the condensed liquid water is able to mix with KSCN to initiate the cooling process. Remarkably, the rapid temperature drop for the KSCN–water binary system is observed with the minimal temperature of −8 °C (Fig. 5b). After the dissolution process is completed, the valves between the two tanks are closed to physically separate the sorbent side and solution side and then the stored thermal energy remains intact over time. When thermal energy is needed, the valves can be re-opened to release the stored sorption potential as heat. To simulate a heat use scenario at low temperature, the temperature of solution is maintained at −2 °C before the opening of the valves. A substantial heat release of the sorbent is observed with a rapid temperature rise to 55 °C (Fig. 5b), verifying the superior heat output performance.

For different energy usage scenarios, this system can be flexibly transformed into other modes, such as continuous cooling (Fig. 5c) and continuous heating (Fig. 5d) with the intermittent heat input as the driving energy (Supplementary Note 2). In the continuous cooling mode, the evaporation of water during the regeneration of endothermic salt will produce a cooling effect (Fig. 5c, $Q_{c,2}$), which together with the dissolution refrigeration effect (Fig. 5c, $Q_{c,1}$) can be used for a 24-h all-day cooling scenario. In the continuous heating mode, the condensation heat (Fig. 5d, $Q_{h,1}$) of water vapour from the sorbent and the sorption heat (Fig. 5d, $Q_{h,2}$) enable this cycle for the continuous heating. It should be noted that the formation of salt solution (Fig. 5d, process 2–3)

https://doi.org/10.1038/s41560-026-01992-0

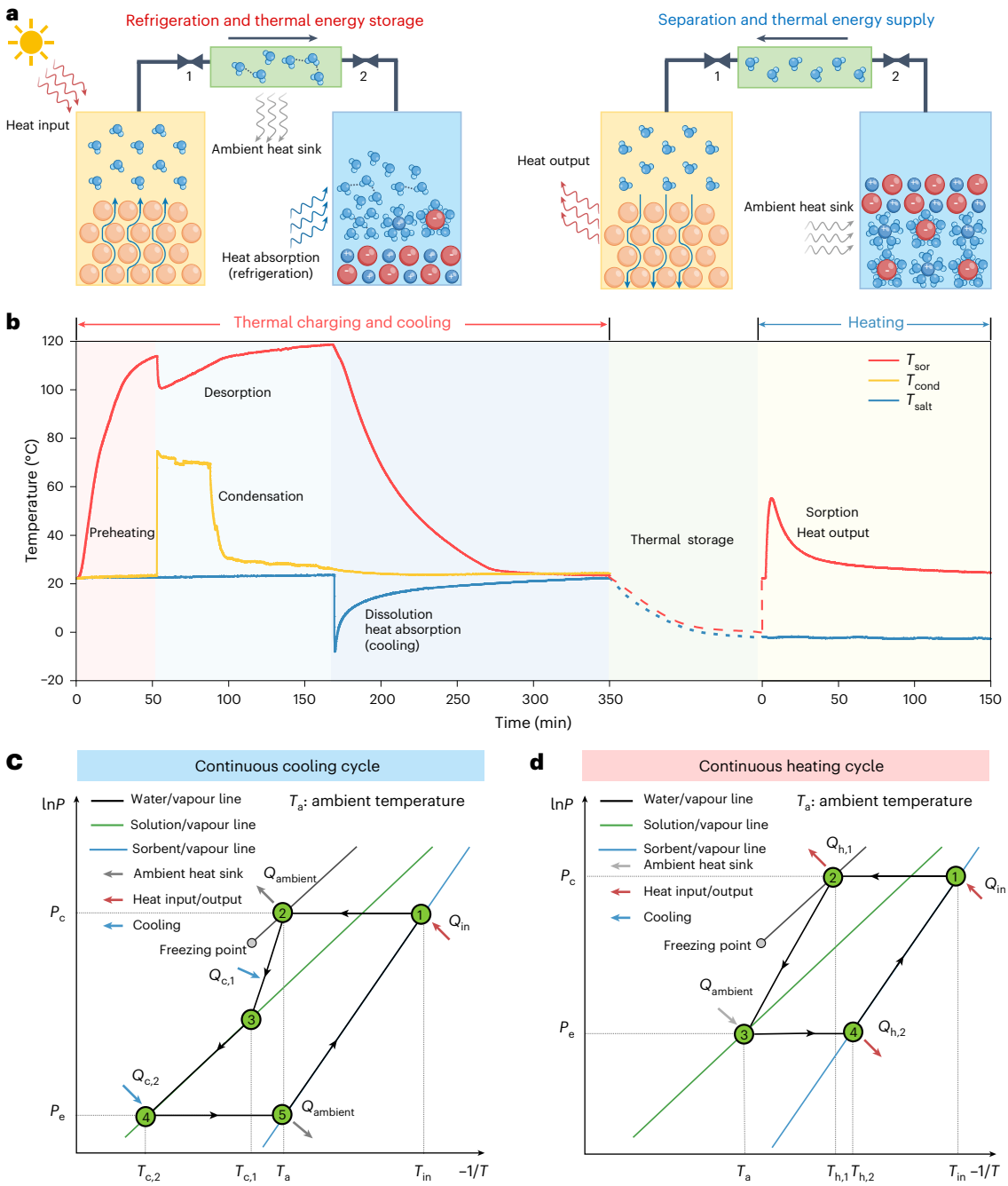

Fig. 5 | Illustration and verification diagram of the cycle and concept of extended applications. a, Schematics of the whole cycle. The yellow, green and blue squares represent the heat storage tank, condenser and refrigeration tank, respectively. The yellow circles represent the sorbent. **b**, The temperature evolutions of three main components over time in the whole cycle. The dashed lines represent the time transition. **c**, The continuous cooling cycle principle. The charge process is depicted by 1–2–3 and the discharge process by 4–5. **d**, The continuous heating cycle principle. The charge process is depicted by 1–2 and the discharge process by 3–4. T_{sor}, temperature of sorbent; T_{cond}, temperature of condenser; T_{salt}, temperature of salt (KSCN); ln$P$, natural logarithm of pressure; P_c, condensation pressure; P_e, evaporation pressure; T_c, cooling temperature; T_h, heating temperature; T_{in}, input temperature.

improves the low-temperature adaptability of this cycle because the freezing point of the salt solution is lower than that of pure water.

For the SdDR system with alternating cooling and heating operation, the COP, as a key metric, can be theoretically calculated to reach 1.13 (Methods). The COPs of other working modes are presented in Supplementary Table 6. Moreover, it should be mentioned that an extra condenser can be added to the basic system configuration (Fig. 5a) to achieve a modified system configuration with more flexible cold output and high cooling power (Supplementary Fig. 21 and Supplementary Note 3). When operated in a high-ratio renewable scenario, the SdDR system has great potential to achieve a levelized cost of thermal energy as low as 0.498 CNY per kilowatt-hour of thermal energy and can reduce CO_2 emissions owing to the zero GWP of all-inclusive materials, showing advantages over the conventional vapour-compression system (Supplementary Notes 4 and 5 and Supplementary Tables 8 and 9).

Conclusions and prospects

The SdDR cycle employs solid–gas sorption to separate the endothermic salt from water while simultaneously recovering the thermal energy, effectively addressing heating and cooling demands that are

misaligned with solar energy availability. A maximum adiabatic temperature change of 37 K has been achieved in the dissolution refrigeration by employing the safe and easily accessible KSCN. Furthermore, the system is capable of reaching a minimum cooling temperature of −25.4 °C in a mere 20 s, rendering it particularly well suited for urgent cooling applications, such as the emergency cooling of data centres and high-power-density electronics. The SdDR system can also deliver heat at 43.2 °C even at a solution evaporation temperature of as low as −13 °C. This shows its suitability even at extremely low ambient temperatures. Compared with previously reported caloric-effect refrigeration technologies using either high applied field strengths or high-grade driving energy (that is, electricity), the SdDR cycle can be powered solely by medium/low-grade heat at a temperature of 80–150 °C.

Although the SdDR cycle and system demonstrate advantages in terms of an extended operation temperature range and multiple working modes, there remain some limitations to be addressed. First, the good heat and mass transfer performance of the system is crucial to achieve a high overall power density. On the sorption side, the sorbent should possess both high thermal conductivity and high vapour diffusivity; however, these two properties typically exhibit a trade-off. Balancing the heat and mass transfer performance of the sorbent is a long-term challenge. From the perspective of thermodynamics, developing novel high-performance vapour sorbents (such as metal–organic frameworks, hybrid sorbents and so on) with low vapour desorption temperatures and high vapour adsorption quantities represents another important direction to exploit the low-grade heat energy and boost the energy storage density. On the salt solution side, active enhancement strategies, such as ultrasonic vibration and falling film evaporation, should be used to accelerate dissolution and evaporative crystallization. In addition, the system operates at low vapour pressures, so maintaining vacuum conditions and stability under pressure fluctuations is also a crucial challenge. New sealing mechanisms need to be investigated to address this issue. Corrosion is another concern due to the existence of aqueous salt solution, and the heat exchanger and envelope of the tank should use corrosion-resistant materials (for example, stainless steel) or coatings to ensure long-term durability. Last but not least, cyclic stability should be further assessed at both material and device levels under diverse operating conditions. In terms of applications, this system shows promise in building energy management, industrial waste heat utilization and thermal management in data centres, among others.

Methods

Preparation of composite sorbents

Zeolite 13X was initially dried at 200 °C. The dried zeolite 13X was then impregnated with a 30 wt.% LiCl solution under vacuum at 80 °C for 3 days. After impregnation, the mixture was filtered to separate the LiCl@Zeolite13X composite from the excess salt solution. The composite sorbent was subsequently washed with anhydrous ethanol to remove residual salt solution from the particle surface. Finally, the composite sorbent was dried at 200 °C for 12 h to ensure complete dehydration.

Fabrication of the SdDR system

The SdDR system consisted of a sorption tank, a condenser and a refrigeration tank, primarily constructed from 316 l of stainless steel, except for the copper exchanger in the sorption tank. Vacuum valves were used to connect the components. Additional structural details are provided in Supplementary Figs. 20 and 22.

Drawing the phase diagram of the KSCN–water system

The binary phase equilibria of the KSCN–water system were measured using differential scanning calorimetry. Samples were prepared by mixing KSCN (99.99%) and water to form binary solutions. First, 5 µl of each solution was sealed in aluminium pans. Each sample was then heated to 30 °C at 5 °C min^{-1}, held isothermally for 10 min to ensure equilibrium and then cooled to −60 °C at 5 °C min^{-1}. The point of the phase diagram boundary was determined by the first peak value of each result, while the eutectic temperature was identified from the second peak values. Data for temperatures above 0 °C were obtained from the literature[27] and integrated with the experimental results to construct a complete phase equilibrium profile for the KSCN–water system over the investigated temperature range.

Measuring the temperature drop of dissolution cooling salt

A specific amount of KSCN was placed in a plastic beaker surrounded by insulating cotton to form an adiabatic environment. A thermocouple was positioned inside a plastic cup. Water was added to form a saturated solution and temperature changes were recorded using a data logger. A magnetic stirrer was employed to mix KSCN with water rapidly. More details are shown in Supplementary Fig. 23.

For measuring the temperature drop of the dissolution cooling salt, water at varying temperatures was mixed with the salt. Two temperature sensors were used to record the initial temperature of the salt and water. A certain amount of salt was previously placed in a crystallizing dish and water (precooled to temperatures between 0 °C and 20 °C) was added to form saturated solutions. A magnetic stirrer was employed to mix the solution throughout the process (Supplementary Videos 1–5).

Vapour sorption isothermals of composite sorbents

The vapour sorption isotherms of LiCl@Zeolite13X at different humidity conditions were measured by an adsorption apparatus (ASAP 2020 plus, Micromeritics) under different temperatures.

Thermal energy storage capacity of composite sorbents

The heat flow in the sorption was measured on a thermogravimetric analyser (STA 449 C, Netzsch) equipped with a moisture humidity generator (MHG32, ProUmid). First, the LiCl@Zeolite13X sample was completely dried at 200 °C. Then, the sample was placed inside a thermogravimetric analyser and kept at 30 °C for the given humidity condition (10% and 60%) for 12 h for sorption.

Theoretical calculations for the selection of endothermic materials

Assuming the salt dissolves completely in water, the average specific heat of the solution throughout the process is approximated as that of water. The theoretical temperature drop (ΔT) and cooling energy density (C_{de}) of the endothermic material can be calculated using the following equations:

$$\Delta T = \frac{\Delta H_{dis}}{\left(\frac{1}{S}+1\right)c_p}$$

$$C_{de} = \frac{\Delta H_{dis}}{\left(\frac{1}{S}+1\right)},$$

where ΔH_{dis} is the enthalpy change of dissolution at 25 °C, S is the solubility of the endothermic material at 25 °C and c_p is the specific heat capacity of water. Dissolution enthalpy values were sourced from ref. 28.

Theoretical calculations for the selection of different working pairs

To preliminarily screen high-performance materials, we evaluated the performance of the different pairs of sorbents and endothermic materials.

The cooling power was determined by the water production rate and enthalpy of dissolution

$$P_{cooling} = \frac{P_{pro}}{S\Delta H_{dis}},$$

where $P_{cooling}$ is the cooling power, P_{pro} represents the water production rate and S is the solubility of the endothermic material at 25 °C.

Considering different water desorption rates, the water production rate can be determined by

$$P_{pro} = \frac{P_{des} M_w}{\Delta H_{des}},$$

where P_{des} denotes the desorption power, calculated for different assumed conditions, M_w is the molar mass of water and ΔH_{des} is the enthalpy of desorption[29–34].

The TESD is related to the ratio of sorbent and endothermic materials. We assumed the water can be desorbed completely, allowing the endothermic materials to dissolve and form a saturated solution. The reaction of hydration of hygroscopic salts was selected based on the transition below 150 °C.

$$\mathrm{TESD} = \frac{\Delta H_{des}}{(S + 1 + m_{sor}) M_w},$$

where m_{sor} is the mass ratio of sorbent to water when salt (sorbent) becomes hydrated salt.

Theoretical minimum supply temperature of composite sorbents

The Clausius–Clapeyron equation linking equilibrium pressure and temperature is in the following form at the standard condition:

$$\ln \frac{P}{P_0} = \frac{\Delta H}{RT} - \frac{\Delta S}{R},$$

where P_0 is the standard pressure that equals 101.325 kPa and ΔH, ΔS and R are the standard enthalpy change, entropy change for the reaction and the ideal gas constant, respectively. For the hydrate conversion reaction, the equilibrium temperature was derived from the vapour pressure of the KSCN solution.

Isothermal entropy changes of KSCN–water system

On the basis of dissolution enthalpy data at various temperatures, the value at 30 °C was selected for theoretical calculations.

$$\Delta S_{isothermal} = \frac{\Delta H_{dis}}{\left(1 + \frac{1}{S}\right) T},$$

where $\Delta S_{isothermal}$ is the isothermal entropy change per mass and S and T are the solubility of KSCN and temperature, respectively.

Adiabatic temperature changes as the applied chemical potential

The chemical potential is determined by the following equation:

$$\mu = RT \ln(\gamma x),$$

where μ is the chemical potential and γ and x are the activity coefficient and molar friction of KSCN, respectively.

The activity coefficient is calculated based on the theory of Pitzer[35]

$$\ln \gamma = Z_{K^+} Z_{SCN^-} f + m \frac{2 \nu_{K^+} \nu_{SCN^-}}{\nu} B + m^2 \frac{4 (\nu_{K^+} \nu_{SCN^-})^{\frac{2}{3}}}{3\nu} C^{\gamma},$$

where Z_{K+}, Z_{SCN-}, υ_{k+}, υ_{scn-} and v are the number of charges, the number of ions and the total number of ions, respectively, f and B are functions of ionic strength, and m and C^{γ} are the concentration of ions and pitzer parameter of KSCN.

f can be calculated by

$$f = -A \left[\frac{I^{\frac{1}{2}}}{1 + bI^{\frac{1}{2}}} + \frac{2}{b} \ln(1 + bI^{\frac{1}{2}}) \right],$$

where A is the Debye–Hückel coefficient of the permeability coefficient that is determined by the temperature and the characteristics of the solvent, b is the empirical constant and I is the ionic strength.

Similarly, B can be calculated by

$$B = 2\beta^0 + \beta^1 \frac{2\left[1 - \left(1 + aI^{\frac{1}{2}} - \frac{1}{2}a^2 I\right) e^{-aI^{\frac{1}{2}}}\right]}{a^2 I},$$

where β^0, β^1 and a are the pitzer parameters of KSCN and the empirical parameter, respectively.

The ionic strength can be calculated by

$$I = \frac{1}{2} \sum m_i Z_i.$$

Measurement of dissolution enthalpy

We used calorimetry to measure the dissolution enthalpy (Supplementary Fig. 24). First, a certain amount of KSCN was added to a heat-insulated container, followed by adding water to dissolve the salt at a certain temperature (the mass ratio of water to salt depended on the solubility of the salt at this temperature). Upon reaching the minimum temperature, the heating feedback switch was activated until the solution temperature reverted to its initial temperature at which the solution was saturated. The quantity of electricity consumed for the heating feedback process was calculated and recorded as the dissolution enthalpy.

Measuring the effect of multistage dissolution refrigeration

The cooling capacity of each stage was measured using the heat balance method. First, 100 g of KSCN was added, followed by the addition of 10 g of water at each stage. The heat exchanger, made of copper, was equipped with fine-wire thermocouples (type K) at the inlet and the outlet was placed in a plastic beaker. Similarly, a fine-wire thermocouple was placed in the plastic cup and the entire device was surrounded by insulating material to minimize heat loss (Supplementary Fig. 25). The flow of water through the heat exchanger was controlled by a peristaltic pump. When the temperature of the solution reached its minimum, the peristaltic pump was activated to initiate heat exchange. The cooling capacity of each dissolution was calculated on the basis of the following equation:

$$Q = \sum c_p m_f (T_{in} - T_{out}),$$

where m_f is the water flow rate and T_{in} and T_{out} are the temperature of the water inlet and outlet, respectively.

Measuring the temperatures of the hot and cold ends of the SdDR system

A thermocouple welded to a Klein flange blind plate measured the sorbent temperature (hot end). A cotton tissue, with a high capacity for adsorbing the solution, was fixed to the surface of the heat exchanger of the refrigeration tank, and a thermocouple, also welded to the Klein flange blind plate, was brought into contact with the cotton tissue to measure the temperature of the cold end (Supplementary Fig. 22). This method of measurement was exclusively utilized in Fig. 4a. In the case of other conditions, the cotton tissue was not employed; rather the thermocouple was in direct contact with the solution.

Measurement of sorption thermal storage density

We used calorimetry to measure the sorption thermal storage density. Thermally conductive oil was used as the heat transfer fluid and the

temperature of the inlet and outlet oil in the heat exchanger was collected to calculate the overall thermal storage density.

$$E_{\text{thermal}} = \frac{\sum c_{\text{p,oil}} m_{\text{f,oil}} (T_{\text{in}} - T_{\text{out}})}{m_{\text{LiCl@Zeolite13X}}},$$

where E_{thermal} is the thermal storage density, $c_{\text{p,oil}}$ is the specific heat capacity of thermal conductive oil and m_{f}, T_{in} and T_{out} are the thermal conductive oil flow rate and the temperature of the inlet and outlet in the heat exchanger, respectively.

Calculation of theoretic separation efficiency

According to the sorption isotherms, sorption is a function of temperature and relative vapour pressure; however, the effect of temperature is minimal and can be neglected (Supplementary Fig. 10). For the device, the relative vapour pressure is defined as the ratio of the saturated vapour pressure of the solution at a given temperature to the saturated vapour pressure of water vapour at the temperature on the sorption side. The sorption amount (x_0) is taken as the baseline when the temperature difference between the sorption side and the solution side is 15 °C. When the temperatures of both the sorption side and the solution side change, this results in a change in relative vapour pressure for the sorbent, and the corresponding sorption amount (x_1) is observed. The separation efficiency is given by the ratio of x_0 to x_1.

Calculation of the theoretical COP and heat recovery ratio

Over the whole cycle, the heat of the input (Q_{in}) is used for the sensible heat (Q_{sen}) of materials and the heat of desorption (Q_{de}). Sensible heat can be utilized in short-term thermal storage. A portion of the additional cooling is generated throughout the process, thereby improving the overall COP.

$$Q_{\text{in}} = Q_{\text{sen}} + Q_{\text{de}}$$

$$Q_{\text{sen}} = \int_{T_0}^{T_s} c_{\text{p,c}}(T) m_{\text{c}} \mathrm{d}T,$$

where $c_{\text{p,c}}$ is the specific heat capacity of LiCl·H_2O@Zeolite13X, and m_{c}, T_0 and T_{s} are the mass of LiCl·H_2O@Zeolite13X and the ambient and heat source temperatures, respectively.

The specific heat capacity of the composite material can be calculated theoretically as follows:

$$c_{\text{p,c}}(T) = x_{\text{LiCl·H}_2\text{O}} c_{\text{p,LiCl·H}_2\text{O}}(T) + x_{\text{13X}} c_{\text{p,13X}}(T)$$

$$c_{\text{p,LiCl·H}_2\text{O}}(T) = c_{\text{p,LiCl}}(T) + c_{\text{p,H}_2\text{O}}$$

$$c_{\text{p,13X}}(T) = 828 + 1.16T,$$

where $c_{\text{p,H}_2\text{O}}(T)$, $c_{\text{p,LiCl}}(T)$ and $c_{\text{p,13X}}(T)$ represent the specific heat capacity of LiCl·H_2O, LiCl and zeolite 13X that is related to temperature[36,37], $c_{\text{p,H}_2\text{O}}$ is the group contribution of the crystal water molecule to the heat capacity of the hydrated salt independent of temperature and set to 43.5 J K^{-1} mol^{-1}, and $x_{\text{LiCl·H}_2\text{O}}$ and x_{13X} are the mass fraction of LiCl·H_2O and zeolite 13X in the composite material.

For the cooling capacity (Q_{co}), the mass of KSCN is chosen such that it is just soluble at 25 °C and the water amount is determined based on sorption isotherm data, with the relative vapour pressure corresponding to the sorption side and the solution side at 25 °C

$$Q_{\text{co}} = m_{\text{H}_2\text{O}} S \Delta H_{\text{dis.}}$$

Therefore, over the complete cycle, the effective energy output comprises the sorption heat (Q_{so}) and the cooling produced by KSCN dissolution

$$\text{COP} = \frac{Q_{\text{sen}} + Q_{\text{so}} + Q_{\text{co}}}{Q_{\text{in}}}.$$

Data availability

The data that support the findings of this study are available within the article and its Supplementary Information. Source data are provided with this paper.

References

1. International Energy Agency. *The Future of Cooling: Opportunities for Energy-Efficient Air Conditioning* (IEA, 2018).
2. Qian, S. et al. High-performance multimode elastocaloric cooling system. *Science* **380**, 722–727 (2023).
3. Kitanovski, A. Energy applications of magnetocaloric materials. *Adv. Energy Mater.* **10**, 1903741 (2020).
4. Liu, J., Gottschall, T., Skokov, K. P., Moore, J. D. & Gutfleisch, O. Giant magnetocaloric effect driven by structural transitions. *Nat. Mater.* **11**, 620–626 (2012).
5. Krenke, T. et al. Inverse magnetocaloric effect in ferromagnetic Ni–Mn–Sn alloys. *Nat. Mater.* **4**, 450–454 (2005).
6. Ma, R. et al. Highly efficient electrocaloric cooling with electrostatic actuation. *Science* **357**, 1130–1134 (2017).
7. Qian, X. et al. High-entropy polymer produces a giant electrocaloric effect at low fields. *Nature* **600**, 664–669 (2021).
8. Meng, Y. et al. A cascade electrocaloric cooling device for large temperature lift. *Nat. Energy* **5**, 996–1002 (2020).
9. Tušek, J. et al. A regenerative elastocaloric heat pump. *Nat. Energy* **1**, 16134 (2016).
10. Cong, D. et al. Colossal elastocaloric effect in ferroelastic Ni–Mn–Ti Alloys. *Phys. Rev. Lett.* **122**, 255703 (2019).
11. Zhou, G. et al. A multi-material cascade elastocaloric cooling device for large temperature lift. *Nat. Energy* **9**, 862–870 (2024).
12. Mañosa, L. et al. Giant solid-state barocaloric effect in the Ni–Mn–In magnetic shape-memory alloy. *Nat. Mater.* **9**, 478–481 (2010).
13. Lloveras, P. et al. Colossal barocaloric effects near room temperature in plastic crystals of neopentylglycol. *Nat. Commun.* **10**, 1803 (2019).
14. Li, B. et al. Colossal barocaloric effects in plastic crystals. *Nature* **567**, 506–510 (2019).
15. Wu, S. & Li, T. Taking control of salts. *Nat. Energy* **8**, 226–227 (2023).
16. Lilley, D. & Prasher, R. Ionocaloric refrigeration cycle. *Science* **378**, 1344–1348 (2022).
17. Wang, W. et al. Conversion and storage of solar energy for cooling. *Energy Environ. Sci.* **15**, 136–145 (2022).
18. Kim, S., Park, J. H., Lee, J. W., Kim, Y. & Kang, Y. T. Self-recovering passive cooling utilizing endothermic reaction of NH_4NO_3/H_2O driven by water sorption for photovoltaic cell. *Nat. Commun.* **14**, 2374 (2023).
19. Kim, S. et al. Passive isothermal film with self-switchable radiative cooling-driven water sorption layer for arid climate applications. *Nat. Commun.* **15**, 8000 (2024).
20. Wesis, L. B. *Ice Cream, A Global History* (Reaktion, 2011).
21. Kim, S. et al. Liquid-state dipolarcaloric refrigeration cycle with nitrate-based salts. *Science* **390**, eadz7967 (2025).
22. Xu, J. et al. Efficient solar-driven water harvesting from arid air with metal–organic frameworks modified by hygroscopic salt. *Angew. Chem. Int. Ed.* **59**, 5202–5210 (2020).
23. Yan, T. et al. Ultrahigh-energy-density sorption thermal battery enabled by graphene aerogel-based composite sorbents for thermal energy harvesting from air. *ACS Energy Lett.* **6**, 1795–1802 (2021).
24. Xu, J. et al. Sustainable moisture energy. *Nat. Rev. Mater.* **9**, 722–737 (2024).
25. Yue, Y., Gai, W. & Boustras, G. Exploration of the causes of ammonium nitrate explosions: statistics and analysis of accidents over the past 100 years. *Saf. Sci.* **158**, 105954 (2023).

26. Hou, J., Gai, W., Cheng, W. & Deng, Y. Hazardous chemical leakage accidents and emergency evacuation response from 2009 to 2018 in China: a review. *Saf. Sci.* **135**, 105101 (2021).
27. Agrawal, Y. K., Sabbagh, R., Sanders, S. & Nobes, D. S. Measuring the refractive index, density, viscosity, pH, and surface tension of potassium thiocyanate (KSCN) solutions for refractive index matching in flow experiments. *J. Chem. Eng. Data* **63**, 1275–1285 (2018).
28. Lide, D. R. *CRC Handbook of Chemistry and Physics* (CRC, 2010).
29. Donkers, P. A. J., Sögütoglu, L. C., Huinink, H. P., Fischer, H. R. & Adan, O. C. G. A review of salt hydrates for seasonal heat storage in domestic applications. *Appl. Energy* **199**, 45–68 (2017).
30. Blijlevens, M. A. R. et al. A study of the hydration and dehydration transitions of $SrCl_2$ hydrates for use in heat storage. *Sol. Energy Mater. Sol. Cells* **242**, 111770 (2022).
31. Lahmidi, H., Mauran, S. & Goetz, V. Definition, test and simulation of a thermochemical storage process adapted to solar thermal systems. *Sol. Energy* **80**, 883–893 (2006).
32. Iyimen-Schwarz, Z. & Lechner, M. D. Energiespeicherung durch chemische reaktionen. I. DSC-messungen zur quantitativen verfolgung der enthalpieänderungen von speicherstoffen für die hin- und rückreaktion. *Thermochim. Acta* **68**, 349–361 (1983).
33. Grevel, K.-D. & Majzlan, J. Internally consistent thermodynamic data for magnesium sulfate hydrates. *Geochim. Cosmochim. Acta* **73**, 6805–6815 (2009).
34. Ferchaud, C., Zondag, H., Veldhuis, J. & Boer, R. D. Study of the reversible water vapour sorption process of $MgSO_4$ $7H_2O$ and $MgCl_2$ $6H_2O$ under the conditions of seasonal solar heat storage. *J. Phys. Conf. Ser.* **395**, 012069 (2012).
35. Pitzer, K. S. *Activity Coefficients in Electrolyte Solutions* (CRC, 1991).
36. Li, D. et al. Phase diagrams and thermochemical modeling of salt lake brine systems. I. LiCl + H_2O system. *Calphad* **51**, 1–12 (2015).
37. Scuiller, E., Dutournié, P., Zbair, M. & Bennici, S. New approach for measuring the specific heat capacity of reactive adsorbents using calorimetry. *J. Chem. Eng. Data* **68**, 1865–11871 (2023).

Acknowledgements
This work is supported by the National Natural Science Foundation of China (grant nos. 52476210 to S.W., 52325601 to T.L. and 52293412 to T.L.).

Author contributions
S.W. conceived of the concept. K.T. and S.W. designed the device, conducted the experiments for device performance characterization and wrote the paper. X.Z., H.S., R.D. and P.W. completed the data visualization. S.W., J.X. and T.L. completed the review and editing. S.W., T.L. and R.W. supervised the project.

Competing interests
The authors declare no competing interests.

Additional information
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41560-026-01992-0.

Correspondence and requests for materials should be addressed to Si Wu, Ruzhu Wang or Tingxian Li.

Peer review information *Nature Energy* thanks the anonymous reviewers for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2026